Optimize hospital capacity planning with length-of-stay prediction at admission

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1 Introduction

Hospital bed congestion and emergency department crowding cause severe operational bottlenecks, including prolonged boarding, ambulance diversions, and cancelled elective surgeries Olshaker and Rathlev [2006]. Effective capacity planning requires coordinating patient admissions with downstream bed availability, yet operational decisions are often hindered by uncertainty in inpatient duration.

Prior operations research addresses this bottleneck primarily through admission-side gating and queue control. Helm et al. [2011] formulated Markov Decision Process models to establish admission thresholds that balance emergency and scheduled patient flows. Yuan and Su [2026] extend this approach to empirical settings, employing dynamic threshold downgrading policies to automate bed allocation and minimize high-risk patient rejections during demand surges.

However, these frameworks typically assume static or aggregate stay durations, overlooking the high clinical heterogeneity of inpatient length of stay (LoS). To address this limitation, we develop machine learning models to predict inpatient LoS right at admission. Using baseline clinical and demographic intake features, our approach replaces naive duration averages with patient-level discharge forecasts to enable proactive bed turnover and dynamic capacity allocation.

2 Data

The dataset we are using is the Statewide Planning and Research Cooperative System (SPARCS) Inpatient De-identified File provided by the New York State Department of Health [2024]. We collected the data programmatically through the Socrata Open Data API (SODA), requesting the full 2024 file in paginated requests of 50,000 rows each, then cached it locally as a parquet file. It contains information on the recorded facilities, inpatients' demographics, admission types, clinical diagnosis and services, and payment details. Publicly available, the data does not contain any protected health information (PHI) under the Health Insurance Portability and Accountability Act (HIPAA), nor does it contain any individually identifiable data element.

2.1 Overview

There are 2,196,737 rows and 33 columns in the raw data. Each row represents a discharge record and each column describes a piece of inpatient information. The column `los` is our target variable derived from the column `length_of_stay`, with the value `120+` (more than 120 days of stay) converted as `120` (120 days of stay). The final dataset we would be training on contains 1,993,483 rows and 15 columns, including 1 target variable and 14 features.

2.2 Features for Training

The 14 features we would be using for model training are grouped into the following:

Group	Features
Facility / Geography	permanent_facility_id
Patient	age_group, gender, race, ethnicity, zip_code
Admission	type_of_admission, emergency_department_indicator
Clinical	ccsr_diagnosis_code, ccsr_procedure_code, apr_medical_surgical
Payer	payment_typology_1, payment_typology_2, payment_typology_3

Table 1: Predictors By Groups.

2.3 Dropped Columns

The rest of the columns are dropped for reasons such as data leakage, lack of relevance, and redundancy, as shown in Table 2. See Appendix A for the detailed explanations.

Columns Dropped	Reason
discharge_year	Constant
patient_disposition, total_charges, total_costs	Leakage by definition
apr_severity_of_illness_code, apr_risk_of_mortality, apr_drg_code	Condition at discharge
facility_name	Duplicate of another column (multi-collinearity)
apr_severity_of_illness, apr_mdc_description, apr_drg_description, ccsr_diagnosis_description, ccsr_procedure_description	1:1 duplicate of a code column (multi-collinearity)
length_of_stay	Replaced by target
birth_weight	Out of scope
hospital_county, health_service_area, operating_certificate_number	Nested under facility (multicollinearity, Cramér's $V \approx 1.0$)
apr_mdc_code	Coarser version of diagnosis (multi-collinearity, $V = 0.9$)

Table 2: Columns Dropped and Reasons

2.4 Dropped Rows

Some rows are also dropped due to usability and/or relevancy issues. Details and explanations are provided in Table 3 on page 3.

Rows	Count	Reason
Redacted facility records	5,295	Deliberate de-identification (mostly abortion-related). Imputing a facility would create a false signal.
Newborn admissions	197,959	A birth in the hospital is a different process from the one being predicted. Set aside as `df_newborn`.
Ungroupable records	381	Severity 0, mortality undetermined and medical/surgical Not Applicable. Not on the ordered scales and cannot be imputed.

Table 3: Rows Dropped and Justifications

2.5 Missing Values

We assign labels based on the underlying reasons for the missingness, as shown in Table 4.

Column	Label	Meaning
`zip_code`	Suppressed	Blanked by SPARCS for small populations, small cells or abortion. Kept separate from `OOS` (Out of State).
`payment_typology_1`	Not recorded	First payer left blank.
`payment_typology_2`	No second payer	No additional payer.
`payment_typology_3`	No third payer	No additional payer.
`ccsr_procedure_code`	No procedure recorded	Almost all on medical stays, none on surgical stays.

Table 4: Labels for Missing Values

3 Exploratory Data Analysis

3.1 Target Variable

We begin our analysis from the target variable. 1.9% of discharges are stays longer than 30 days, so about 98% last 30 days or fewer. Upon that, most of the stays are short. About half of all discharges end within 3 days: 17.7% at 1 day, 18.1% at 2 days, 15.1% at 3 days. The distribution is strongly right-skewed, as shown in Figure 1. This limited representation of long stays may restrict the model's ability to learn patterns associated with longer stays; therefore, resampling techniques might be needed during training or the model's performance should be evaluated with respect to different groups of length of stay.

3.2 Categorical Analysis

We next examine the feature variables to better understand the characteristics of the patient population and admission patterns. The dataset is skewed towards older patients, with 35.5% of admissions involving patients aged 70 or older and another 29.1% involving patients aged from 50 to 69. In contrast, minors aged from 0 to 17 account for only 5.2% of all admissions. Emergency admissions predominate, accounting for 73.1% of admissions, and 68.9% of patients pass through the emergency department. Medicare (46.4%) and Medicaid (26.8%) dominate the primary payer. Just over half (52.5%) have no second payer, and 88.6% of the discharges have no third payer.

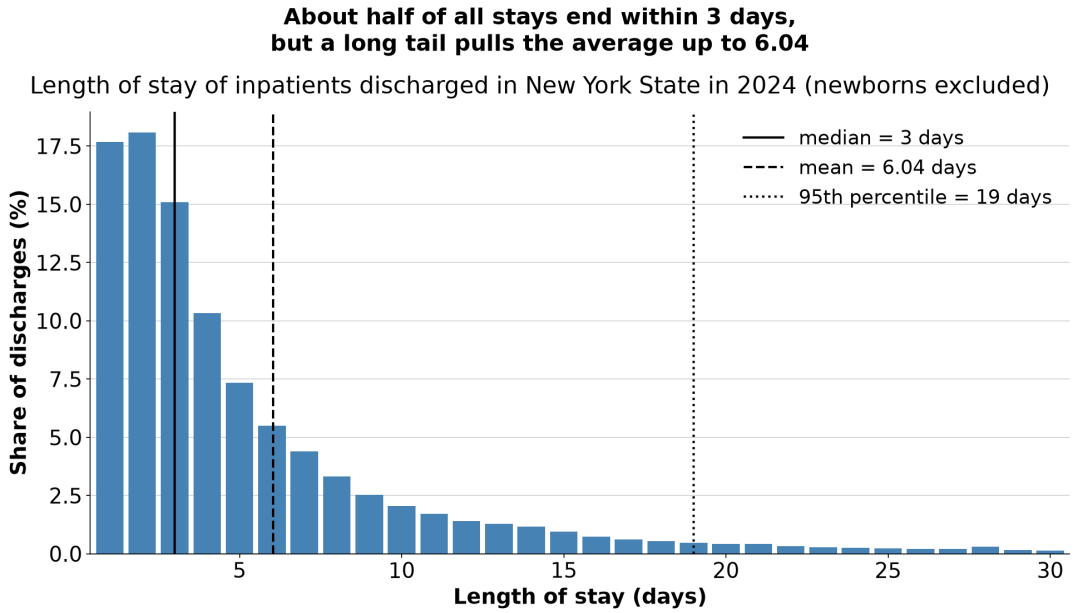


Figure 1: Share of Discharges by Number of Days (Only stays of 1 to 30 days are shown)

3.3 Bivariate Analysis

We then move on to analyze the relationship between each feature and the target variable. In Figure 2, we could see that the diagnosis separates the typical stay strongly from about 2 days to about 13 days, suggesting that `ccsr_diagnosis_code`is a strong predictor for the length of stay.

Moreover, among the 12 most common procedure categories, typical lengths of stay range from approximately 2 days for spontaneous vaginal delivery to 7 days for venous and arterial catheter placement. Delivery-related procedures, including spontaneous vaginal delivery and cesarean section, have the shortest stays (2-3 days), whereas catheter placement, non-invasive ventilation, and blood transfusion are associated with the longest stays (5-7 days), marking `ccsr_procedure_code` as another critical predictor of the length of stay.

We also examined the potential impact of selected features with smaller cardinalities, shown in Figure 3. The distributions of length of stay overlap substantially across all groups, suggesting that no demographic or admission-related characteristic individually distinguish the length of stay particularly well, but predictive bias remains to be evaluated.

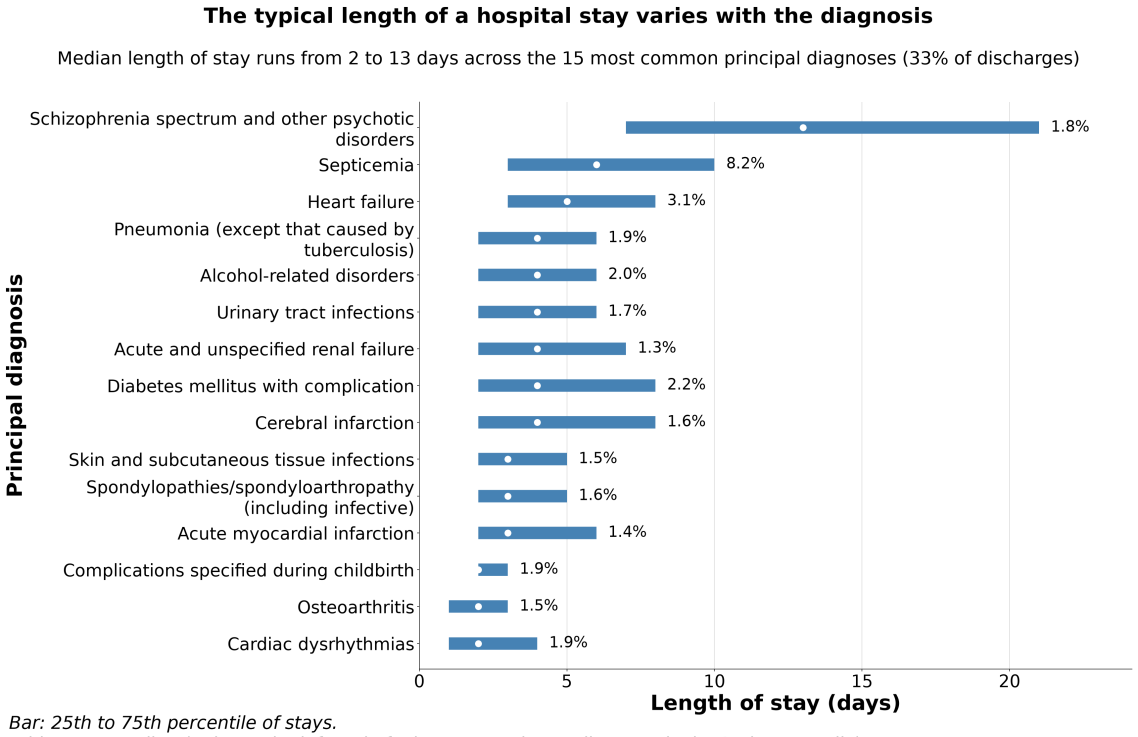


Figure 2: Length of Stay by Principle Diagnosis

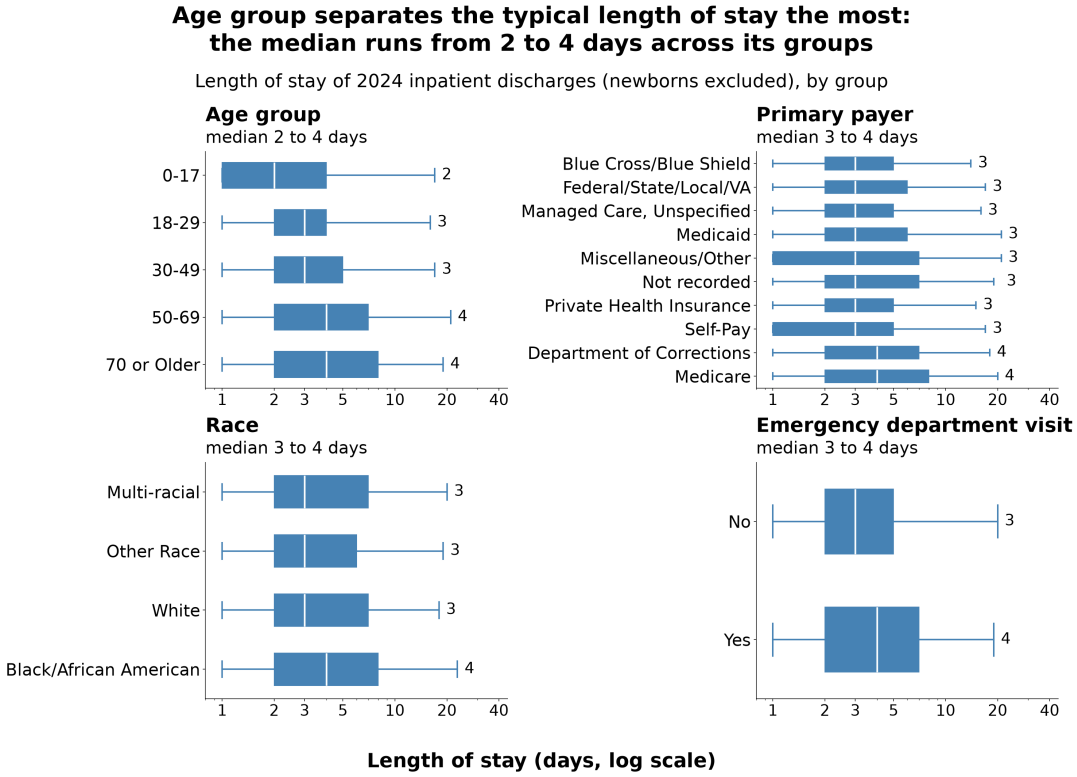


Figure 3: Length of Stay by Selected Inpatient Groups

4 Problem description and potential impact

4.1 Problem description

We model the prediction of inpatient LoS as a **supervised regression task**. Formally, each patient admission is treated as an independent and identically distributed observation represented by an intake feature vector $x \in \mathcal{X} \subset R^d$ observed at the point of admission. The ground-truth response is a continuous target variable $y \in \mathcal{Y} \subset R^+$ denoting total stay duration in days (measured from admission to discharge). Our goal is to estimate an empirical risk-minimizing hypothesis $f_\theta : \mathcal{X} \to \mathcal{Y}$. While duration can be discretized into categorical intervals for ordinal classification, formulating y as a continuous target preserves fine-grained capacity consumption essential for census forecasting. We also imposed operational constrains that model inference occurs **strictly at admission**. To prevent data leakage, we exclude all post-admission variables specified in Table 2 which are unobservable at intake.

The New York State Department of Health [2024] SPARCS administrative dataset has been widely used in health services research. Chen et al. [2017] utilized SPARCS to analyze spatio-temporal disease clustering and community-level discharge counts. Koltenyuk et al. [2026] modeled post-acute exit pathways, where baseline severity metrics like APR-DRG risk of mortality and illness severity predict discharge disposition. To our knowledge, the dataset has not been leveraged to develop admission-constrained regression models for bed capacity forecasting.

4.2 Potential impact

Our model would substantially improve operational efficiency for the hospital administration team from inpatient bed coordinators to hospital capacity managers. When deployed at the patient intake stage, our model automates stay estimation, outputting an individualized discharge horizon as soon as admission paperwork is logged.

The immediate implications of our model in hospital operation are:

- **Early Bed-Turnover Scheduling:** Operations teams can proactively match expected ward departures against incoming emergency department boarders hours in advance, reducing access block and ambulance diversion.
- **Surge and Staffing Calibration:** Nurse managers can adjust shift staffing ratios dynamically based on forecasted multi-day census rather than uncalibrated headcounts.

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A Detailed Justification for Dropped Columns

Columns Dropped	Reason	Justification
`discharge_year`	Constant	Only one value (2024), so it cannot explain length of stay.
`patient_disposition`, `total_charges`, `total_costs`	Leakage by definition	They describe the completed stay and are not known at admission. Charges and costs grow with every extra day.
`apr_severity_of_illness_code`, `apr_risk_of_mortality`, `apr_drg_code`	Condition at discharge	The 3M APR-DRG Methodology Overview ([3M Health Information Systems, 2023, p. 53]) says they represent the condition at discharge, including conditions that developed during the stay.
`facility_name`	Duplicate of another column (multicollinearity)	Can be determined by `permanent_facility_id`.
`apr_severity_of_illness`, `apr_mdc_description`, `apr_drg_description`, `ccsr_diagnosis_description`, `ccsr_procedure_description`	1:1 duplicate of a code column (multicollinearity)	Same information as the code column, written as text, so each pair is perfectly collinear.
`length_of_stay`	Replaced by target	Replaced by the numeric `los`.
`birth_weight`	Out of scope	Filled only for newborns (99.97% of newborns). Newborns are a separate cohort, so it is blank for everyone else.
`hospital_county`, `health_service_area`, `operating_certificate_number`	Nested under facility (multicollinearity, Cramér's V ≈ 1.0)	`permanent_facility_id` determines all three, so they add no information and would be collinear with it.
`apr_mdc_code`	Coarser version of diagnosis (multicollinearity, $V = 0.9$)	MDC is defined from the principal diagnosis ([3M Health Information Systems, 2023, p. 57]), so `ccsr_diagnosis_code` already carries it and the two are strongly collinear.

Table 5: Columns Dropped and Justifications